# Case 06: Two-Dimensional Firebrand-Driven Spread

## 1 Verification purpose

This case verifies that Eulerian spotting produces a genuinely two-dimensional, firebrand-driven fireline under a manufactured two-component, space–time wind. It exercises emission, vector transport, deposition, direct spot ignition, and level-set propagation together.

The test exposes loss of either wind component, incorrect direction conversion, one-dimensional source geometry, missing secondary ignitions, non-monotone burned area, and stale transient output. Success requires nested perimeters near 30, 60, 90, and 120 s that advance in both directions and exceed the surface-only reach.

## 2 Mathematical formulation and numerical implementation

The prescribed horizontal wind is

$$u(x,t) = \bar{u} + A\sin(2\pi x/\lambda_x)\sin(2\pi t/T), \tag{1}$$
$$v(y,t) = \bar{v} + B\sin(2\pi y/\lambda_y)\sin(2\pi t/T), \tag{2}$$

with $\bar{u} = 6.71\,\mathrm{m\,s^{-1}}$, $A = 3.355\,\mathrm{m\,s^{-1}}$, $\bar{v} = 1.0\,\mathrm{m\,s^{-1}}$, $B = 0.5\,\mathrm{m\,s^{-1}}$, $\lambda_x = \lambda_y = 100\,\mathrm{m}$, and $T = 60\,\mathrm{s}$. The preprocessor evaluates these equations at every cell centre and every one-second meteorology time. It writes $|\mathbf{u}|$ in mph and converts the transport-to angle to ELMFIRE meteorological wind-from convention.

For each Eulerian source tracker, the current spotting implementation advances

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{u}(\mathbf{x}_k, t_k)\Delta t_k, \qquad \Delta t_k = \Delta x/\max(|\mathbf{u}|, 0.01),$$

and distributes the source ember count through the configured downwind lognormal landing distribution. Direct ignition updates the level-set state, after which both the primary headfire and spot fires create new burning cells. The configured surface reference speed is $R_s = 0.71\,\mathrm{m\,s^{-1}}$ [1], and the corresponding emission residence time is $t_e = \Delta x/R_s = 14.08\,\mathrm{s}$.

The prescribed $u$ and $v$ components determine an individual transport characteristic, but they do not uniquely determine the maximum coordinates of the coupled fireline in this configuration. The observed $\phi = 0$ perimeter is the union of the surface-propagated primary fire and surface-propagated spot fires created after nonlocal, probability-distributed deposition. Consequently, $x_{\max}$ and $y_{\max}$ do not generally satisfy the uncoupled equations $dx_{\max}/dt = u(x_{\max}, t)$ and $dy_{\max}/dt = v(y_{\max}, t)$. An exact endpoint metric would require an independent solution of the complete coupled problem.

## 3 Derivation of expected behavior

Because the wind has positive mean components, successive fronts must progress downwind and develop a nonzero crosswind span. The level-set burned set is irreversible, so for burned masks $B_j = \{(x,y) : \phi(x,y,t_j) \le 0\}$ the expected inclusion is $B_j \subseteq B_{j+1}$. A conservative surface-only leading-edge bound is

$$x_{\mathrm{surf}}(120) = x_0 + R_s(120) = 90.2\,\mathrm{m}.$$

A front substantially beyond this location, together with nonzero ember-ignition cells, demonstrates that firebrands rather than the surface headfire alone control the long-range advance. Accordingly, the

principal expected result is a family of nested two-dimensional contours rather than a one-dimensional time of arrival (TOA) curve.

The largest generated transport speed is

$$|\mathbf{u}|_{\max} = \sqrt{(6.71 + 3.355)^2 + (1.0 + 0.5)^2} = 10.1762 \text{ m s}^{-1},$$

so preprocessing sets $\Delta t = 0.5(10)/10.1762 = 0.491344$ s. The surface residence time is independently

$$t_e = 10/0.71 = 14.0845 \text{ s}.$$

Starting from $x_0 = 5$ m, a surface-only head fire would be at

| $t$ (s) | 30 | 60 | 90 | 120 |
|---|---|---|---|---|
| $x_{\text{surf}}$ (m) | 26.3 | 47.6 | 68.9 | 90.2 |

and would advance $0.71(30) = 21.3$ m between successive target times. The declared 20-m interval threshold is therefore slightly below the surface-only analytical increment but still requires two resolved cells. At 120 s, the firebrand-excess test $x_{\max} - 90.2 \geq 30$ m is equivalently $x_{\max} \geq 120.2$ m, together with a positive ember-ignition count. Exact irreversible evolution gives $|B_j \setminus B_{j+1}|/|B_j| = 0$; the numerical allowance is 0.02. The final span reference is qualitative nonzero two-dimensional growth, made quantitative as at least $40 \text{ m} = 4\Delta x$.

# 4 Verification metrics and acceptance criteria

The postprocessor selects transient level-set field rasters within 2 s of all four target times and excludes the two-cell halo. It calculates: (i) completeness of all four contours; (ii) the fraction $|B_j \setminus B_{j+1}|/|B_j|$, limited to 0.02; (iii) downwind leading-edge advance, required to be at least 20 m in every interval; (iv) final crosswind span, required to be at least 40 m; and (v) firebrand excess advance $x_{\max}(120) - x_{\text{surf}}(120)$, required to be at least 30 m with at least one ember-ignition cell. The overall result is PASS only when every component passes.

## 4.1 Justification of metric quantification

The thresholds are expressed in grid-resolvable units. Selecting a contour within 2 s limits time mismatch to less than 7% of the earliest 30 s target and only a few solver steps. Exact level-set evolution is nested, so the 2% burned-area allowance is reserved for zero-contour classification along a cell-wide perimeter; a larger loss would indicate physical unburning or inconsistent dump selection. The minimum interval advance of 20 m, final crosswind span of 40 m, and firebrand excess of 30 m equal two, four, and three cells, respectively. These multi-cell thresholds cannot be satisfied by a stationary front, a single misclassified boundary cell, or the surface-only reference, yet they do not prescribe a detailed perimeter shape that the manufactured problem does not supply. At least one ember-ignition cell is an exact mechanism check, and all four times are required to support the claimed two-dimensional evolution.

## 4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

| Metric | Output, region, and aggregation | Acceptance criterion |
|---|---|---|
| Perimeter completeness | Unique transient level-set field raster selected within 2 s of 30, 60, 90, and 120 s. | 4 of 4 target times |
| Nesting violation | Maximum $\lvert B_j \setminus B_{j+1}\rvert/\lvert B_j\rvert$ over consecutive physical burned masks. | $\leq 0.02$ |
| Downwind progression | Increase in maximum burned-cell $x$ between consecutive target times. | $\geq 20$ m every interval |
| Final crosswind span | $\max(y) - \min(y)$ of the final physical burned mask. | $\geq 40$ m |
| Firebrand-driven excess | $x_{max}(120) - [x_0 + R_s(120)]$, together with the final ember-ignition map. | $\geq$ 30 m and at least one ember-ignited cell |
| Overall decision | Conjunction of completeness, nesting, progression, span, and excess-advance checks. | PASS only if all pass |

## 5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the base two-dimensional run. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

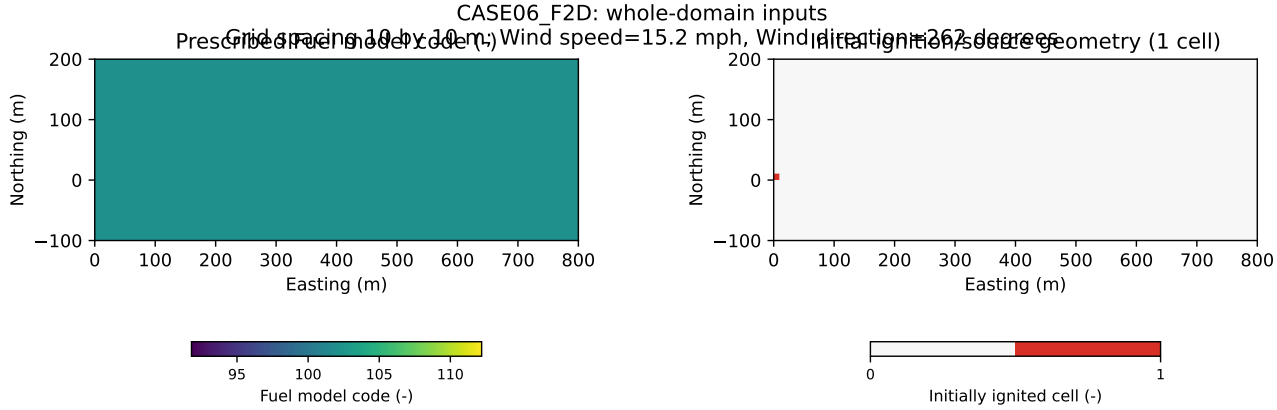



Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Before inspecting results, the predicted output is four nested contours elongated in positive $x$, displaced and broadened in $y$, with many secondary ember ignitions and a final leading edge far beyond 90.2 m.

Table 2: Configured two-dimensional transport case experiment.

| Parameter | Value | Role |
|---|---|---|
| Physical domain | $800 \times 300$ m | two-dimensional propagation |
| Cell size | 10 m | reference spatial resolution |
| Raster grid | $84 \times 34$ cells | includes two-cell halo |
| Meteorology | 121 bands at 1 s | space–time wind |
| Simulation step | 0.491 026 s | wind CFL approximately 0.5 |
| Initial ignition | $(5, 5)$ m | localized upwind source |
| Surface headfire | enabled | primary moving source |
| Generation | PER-AREA, 10 pcs m$^{-2}$ s$^{-1}$ | deterministic ember supply |
| Accumulation / ignition | Eulerian / direct | tested physical and numerical processes |
| Crosswind PDF | $\mu_y = 0$, $\sigma_y = 0.1$ m | narrow $k_{\max} = 0$ limit |
| Output times | 30, 60, 90, 120 s | transient perimeter comparison |

## 6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from the base two-dimensional run. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. Counts are divided by physical cell area, giving pcs m$^{-2}$ and exposing the full lateral as well as downwind distribution.

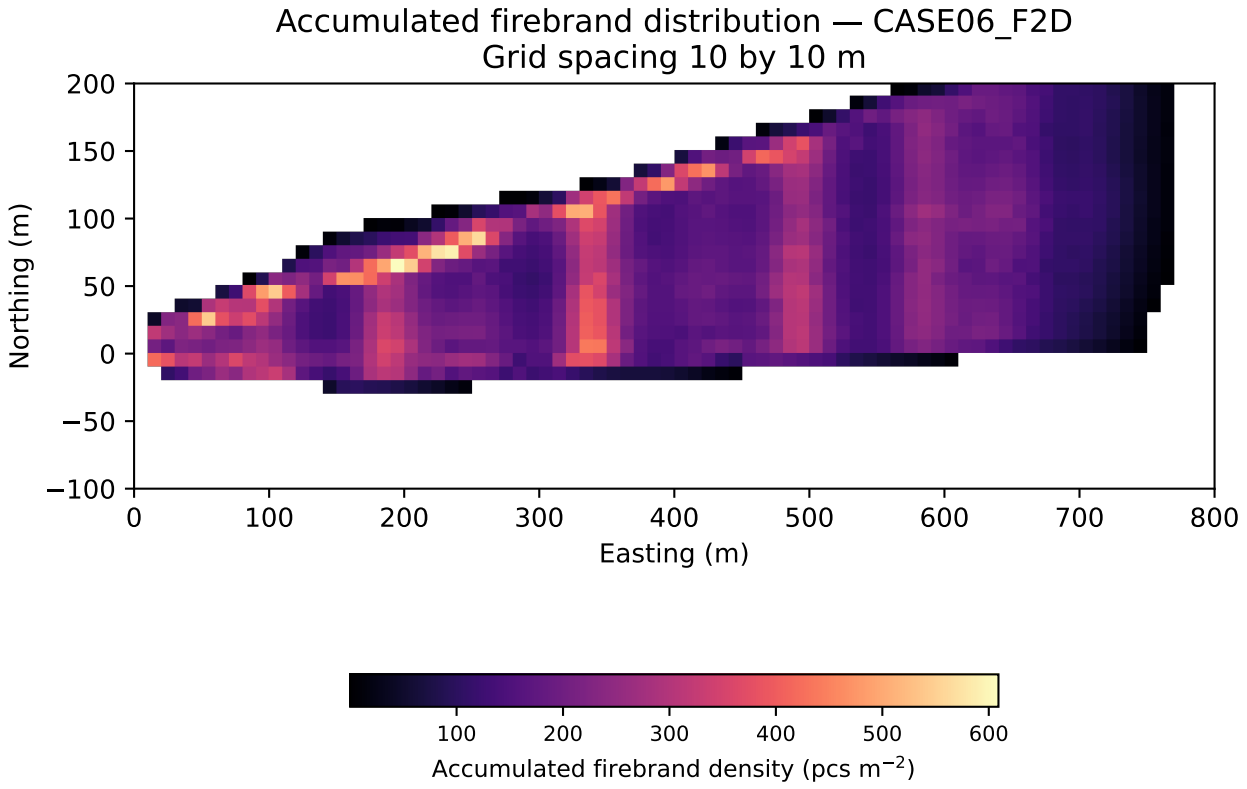



Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

The selected transient perimeter times were 30.44, 60.4, 90.35, 120 s. The identities of the selected fields are retained with the calculated metrics. Figure 3 first presents the evolving zero-level contours, which are the principal two-dimensional reference observable.

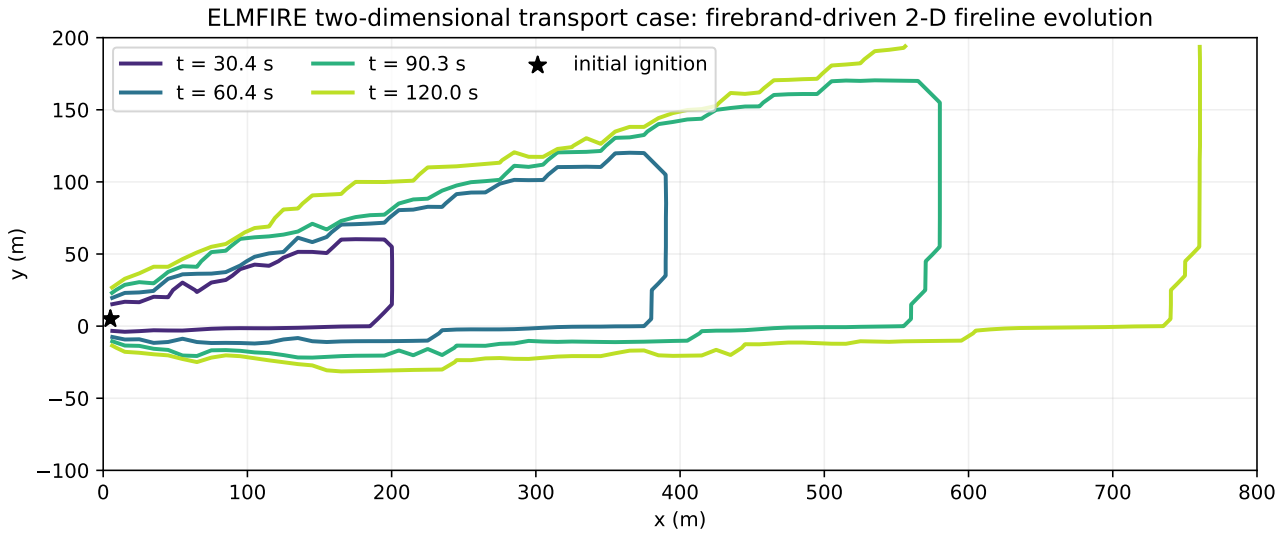



Figure 3: Actual ELMFIRE fire perimeters at four output times under the prescribed two-component, space–time wind. The star marks the initial ignition.

Figure 4 identifies the cells actually ignited by deposited firebrands. The accumulated-firebrand field is not repeated here because its whole-domain distribution is already presented in Figure 2.

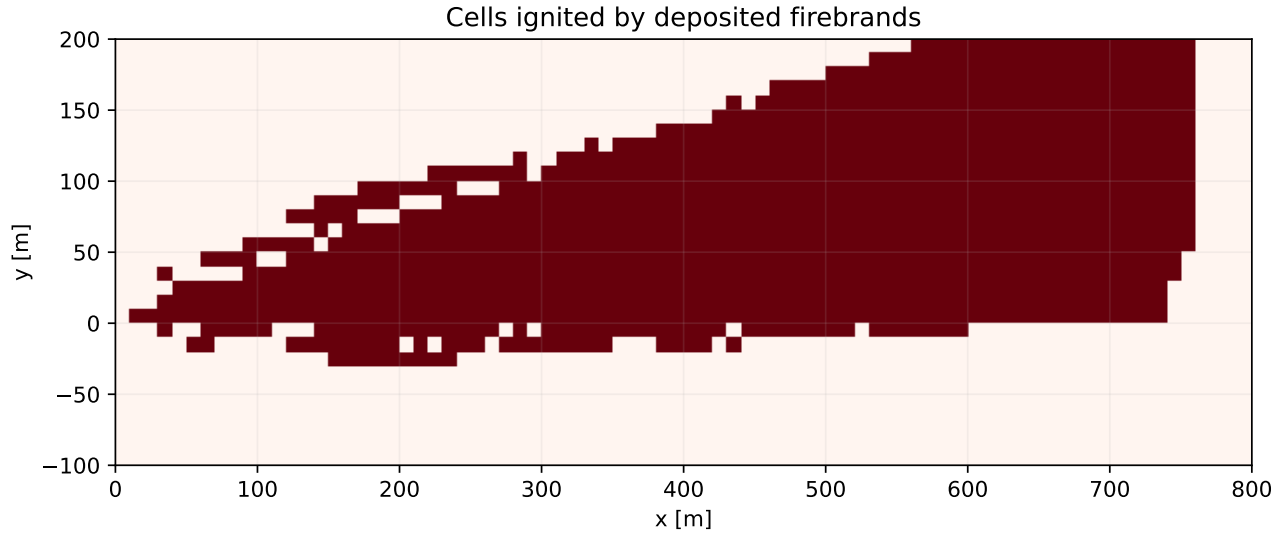



Figure 4: Whole-domain cells marked by ELMFIRE as ignitions caused by deposited firebrands. The numerical halo is excluded.

## 7 Calculated metrics and verification decision

Table 3: Acceptance metrics calculated from actual ELMFIRE outputs.

| Metric | Acceptance limit | Calculated value | Status |
|---|---|---|---|
| Required contours | 4 within 2 s | 30.44, 60.4, 90.35, 120 | PASS |
| Nesting violation | $\leq 0.02$ | 0.0 | PASS |
| Interval advances | each $\geq 20$ m | 190, 190, 180 m | PASS |
| Final crosswind span | $\geq 40$ m | 220.0 m | PASS |
| Firebrand excess | $\geq 30$ m and ignitions | 664.8 m; 1097 cells | PASS |
| Overall | all components true | PASS | **PASS** |

## 8 Technical interpretation

The result is **PASS**. The maximum nesting violation is 0.0, while the leading edge advances by 190, 190, 180 m over the three intervals. The final crosswind span is 220.0 m, confirming that the calculation is not a replicated one-dimensional transect. The final front exceeds the surface-only bound by 664.8 m and the ember-ignition raster contains 1097 positive cells; together these observations attribute the long-range evolving fireline to secondary ignitions from transported firebrands.

## 9 Scope and limitations

This test verifies quantified mechanism and evolution invariants of the current Eulerian 2-D coupling and reconstructs the intended contour diagnostic. The known wind components alone are not treated as an exact outer-fireline solution because surface spread and nonlocal secondary ignitions remain active. The test does not reproduce the comparison solution cell-for-cell, compare Eulerian and Lagrangian contours, validate the landing PDF against experiments, or prove grid convergence. The configured flat terrain isolates horizontal transport; additional terrain-slope cases would be required to verify topographic coupling.

## References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.